

# Slide 1: Model similarity depends on the measurement 0:20

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## Say

- Open plainly: "The point of this talk is that model similarity is not one thing."
- A similarity score is always answering a measurement question: which stimuli, which layer, which behavior, which failure mode?
- The Re-Align challenge is the anchor case because it fixes the metric, CKA, and lets us stress test what that metric actually sees.

## Point at / emphasize

- Title only. Let it breathe; do not preview every project yet.

## Transition

- Move from the headline to the simplest visual intuition: the same metric changes when the stimulus set changes.

### Say

- CKA compares the geometry two models induce over the same set of stimuli.
- If the set contains dogs, cars, birds, and buildings, a lot of the geometry is already category structure. Any competent classifier will agree on much of that.
- If the set is only dogs, the easy category contrast is gone. Now the comparison is about pose, texture, crop, scene, silhouette, and background.
- That is the theme: the measurement does not just compare models; it also inherits the structure of the stimulus set.

### Point at / emphasize

- Point left for broad category clusters, then right for dog-only residual axes.

### Transition

- Now give the one-slide math explanation for why this is not just an intuition.

### Say

- Write the representation as  $X = M + W$ .  $M$  is the between-class part,  $W$  is the within-class deviation.
- CKA operates on centered Gram matrices, so a common misconception is that centering removes the category effect. It does not.
- With many classes, the  $HMM^T H$  term is large: it encodes where class centroids sit relative to one another.
- With one superclass,  $M$  is just one mean point. After centering, it vanishes. What remains is  $HWW^T H$ .
- So the same CKA formula answers two different questions depending on the stimulus set.

### Point at / emphasize

- Point to  $X = M + W$ , then to the  $k = 1$  line. The key phrase is: "centering kills the single mean, not multi-class structure."

### Transition

- That gives us a way to understand the competition as an experimental design over CKA.

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### Say

- The competition fixes the scoring rule: mean pairwise CKA over hidden embeddings.
- But the final evaluator controls the models, layers, preprocessing, and image resolutions. We do not get to optimize directly against the final object.
- So the real challenge is transfer: can a story about what CKA measures survive the move from proxies to the hidden evaluator?
- The submissions themselves are intentionally small: JSON files with either model-layer pairs or image IDs.

### Point at / emphasize

- Use the three right-hand boxes: fixed, hidden, submitted.

### Transition

- There are two tracks. First, the Blue Team task, where the lever is model choice.

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## Say

- Blue controls the models. The images and evaluator are fixed.
- The input is a registry of 141 vision models, each with a representation layer.
- The decision is to choose 20 model-layer pairs.
- Mathematically, this is a densest-subgraph problem: models are nodes, pairwise CKA values are edge weights, and we want the densest 20-node subgraph.

## Point at / emphasize

- Trace input -> choice -> score across the slide.

## Transition

- The Red task is the dual: the models are fixed, but the stimulus set is the lever.

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### Say

- Red controls the images. The model set is fixed by the evaluator.
- The decision is exactly 1,000 ImageNet validation image IDs.
- The score is 1 minus mean CKA, so high score means the selected images make the model representations diverge.
- This turns stimulus selection into a measurement intervention: choose images that remove shared structure and expose disagreement.

### Point at / emphasize

- Trace input -> choice -> score again; contrast it with Blue.

### Transition

- Now show the evidence that broad category structure survives centering, and dog-only images remove it.

# Slide 7: Centering does not neutralize stimuli

1:00

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## Say

- This figure uses real ResNet50 embeddings.
- The top row is mixed ImageNet classes. You can still see the block skeleton after centering.
- The bottom row is dog breeds only. The broad category blocks are gone, and what remains is residual within-superclass geometry.
- This is the visual version of the math: centering is not a magic fix for category structure.

## Point at / emphasize

- Top row: block structure before/after centering. Bottom row: dog-only residual structure.

## Transition

- Given that, the Blue solution should look for models that agree even beyond broad category structure.

### Say

- The Blue strategy treats the proxy CKA matrix as a weighted graph and searches for a dense 20-model clique.
- The selected set concentrates in the attention family: 18 of the 20 submitted models are attention-based architectures.
- The heatmap shows why. The warm block is the high-CKA attention-family region; cooler off-diagonal blocks are weaker cross-family alignment.
- Interpretation: when stimuli are broad, architecture family becomes a strong lens for measured similarity.

### Point at / emphasize

- Point to the bright top-left block and the family separators.

### Transition

- The Red strategy asks the opposite question: how do we choose stimuli so that architecture differences show up?

### Say

- Dogs are not special because we care about dogs. They are useful because they remove the easy contrast.
- The candidate pool was 125 dog and canid synsets, giving 6,250 candidate images.
- We filtered by proxy-model divergence and then optimized the final 1,000 with simulated annealing.
- Inside dogs, the axes that remain are exactly where model families differ: texture, silhouette, crop, scene, pose, and background.

### Point at / emphasize

- Emphasize 125 -> 6,250 -> 5,000 -> 1,000 as the selection funnel.

### Transition

- Then state the result, but keep it tied to measurement rather than leaderboard performance.

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### Say

- On the hidden evaluator, the Red submission scored 0.547 versus the baseline 0.479.
- The proxy score was higher, 0.584, so there is a proxy gap. But the direction transferred.
- For Blue, the submitted set is overwhelmingly attention-family models.
- The scientific claim is not just that the submissions did well. It is that CKA should be reported as conditional on the stimulus set and model pool.

### Point at / emphasize

- Use the numbers quickly; do not get stuck in leaderboard details.

### Transition

- Now broaden the same idea: different probes reveal different kinds of similarity in other model classes.

## Say

- In speech models, the measurement question is not only what representation exists, but when information becomes usable across depth.
- The same utterances can organize by background language at one depth and by utterance identity at another.
- Different encoders can solve the same task while using different processing schedules.
- So this is a depth lens on similarity: where does the model make the relevant variable linearly or geometrically available?

## Point at / emphasize

- Point across columns by depth, then compare Whisper and Canary rows.

## Transition

- Next, replace activation geometry with state-space trajectory geometry.

### Say

- Here the object is not a layer activation; it is a trajectory through quantum state space.
- Shared and decoupled parameterizations trace visibly different paths on the Bloch sphere.
- That gives another definition of processing strategy: not just final output, but the path the system takes to compute it.
- The broader point is the same: choose the right geometry for the question you want to ask.

### Point at / emphasize

- Contrast the left and right Bloch sphere panels: shared parameters versus decoupled parameters.

### Transition

- The last language slides move from internal geometry to behavior under damage.

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### Say

- A different lens is failure behavior. Instead of asking whether activations align, ask whether systems break in the same coordinates.
- TAB puts human aphasia transcripts and model outputs into the same 19-symptom space.
- That lets us ask whether model damage produces human-like deficits, or just superficially similar errors.
- This is useful because failure modes often reveal structure that normal performance hides.

### Point at / emphasize

- Use the right-side panels: common measurement space, failure coordinates, use case.

### Transition

- The lesion slide makes that concrete by separating attention and FFN damage.

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### Say

- Lesions turn architecture into causal evidence. If two components support different computations, damaging them should move behavior in different directions.
- The figure shows raw symptom rates and mean-centered contributions across components.
- Attention and FFN damage produce distinct, depth-stratified failure regimes.
- The important nuance: the symptoms are recognizable, but the mixture is machine-specific. It is not a clean copy of a classical human syndrome.

### Point at / emphasize

- Point left for raw rates, right for mean-centered component contributions.

### Transition

- Close by returning to the measurement claim, now across all the examples.

## Say

- The throughline is that similarity depends on the measurement lens.
- Stimuli can inflate or remove agreement. Depth tells us when information becomes usable. State trajectories reveal computational routes. Lesions reveal causal failure coordinates.
- So the right question is not "are these models aligned?" in the abstract.
- The right question is: aligned with respect to which probe, which stimuli, and which processing strategy?

## Point at / emphasize

- Move down the three boxes: stimuli, depth, state and damage.

## Transition

- End cleanly: "That is the measurement discipline I think representational alignment needs."